Metal Detector



Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Door Frame Metal Detector (DFMD); Hand Held Metal Detector (HHMD)

What it is used for

A metal detector identifies metal objects using electromagnetic sensing, either through a walk-through frame or a handheld scanner. Operate it according to the security procedure, check its function, and conduct screening respectfully and consistently.

Main parts

- Walk-through detector frame or handheld detector body
- Detection coils or sensing panels
- Control unit and status display
- Sensitivity and alarm settings
- Power supply, battery, or charging connection
- Audible alarm and indicator lights

Before you start

1. Confirm the detector type and follow the site's screening procedure.
2. Inspect the frame or handheld unit, cable, battery, and controls for damage.
3. Check the battery or power supply and confirm the unit starts correctly.
4. Perform the required function check using the approved test method.
5. Set sensitivity and alarm parameters according to authorised site settings.
6. Ensure the screening area is organised, accessible, and clearly marked.

How to use

1. Switch on the detector and allow any required self-test to finish.
2. Verify that the ready indicator or status display shows normal operation.
3. For a walk-through frame, direct the person to remove items as required by site policy.
4. Ask the person to walk through the frame at the specified pace and direction.
5. Observe alarm indicators and follow the site's secondary-screening procedure if an alarm occurs.
6. For a handheld detector, explain the process and maintain respectful screening distance.
7. Move the handheld unit slowly and evenly around the approved screening areas without touching the person unnecessarily.
8. Keep the detector at the recommended distance from the body or item being checked.
9. If an alarm occurs, follow authorised procedures to identify the source and resolve the alert.
10. Repeat the check when required by the screening protocol.
11. Record equipment faults or screening incidents according to site policy.
12. Switch off the device when no longer required and store it securely.

⚠ Safety

- Only trained and authorised personnel should conduct security screening.
- Follow privacy, consent, accessibility, and respectful-treatment requirements in site policy.
- Do not use a faulty detector or ignore failed function checks.
- Do not touch or search a person beyond the authority and procedure permitted at the site.
- Keep handheld detectors away from water and protect them from impact.
- Follow site emergency procedures if a prohibited or dangerous item is suspected.

Care & storage

- Clean the detector surfaces using a compatible approved method.
- Check batteries, charging connections, and cables regularly.
- Perform function checks at the intervals required by site policy.
- Protect sensing panels and handheld coils from impact or moisture.
- Report inconsistent alarms, damaged parts, or repeated faults.

Common mistakes to avoid

- Skipping the required function check before screening.
- Using incorrect sensitivity settings or ignoring site procedures.
- Moving the handheld detector too quickly or too far from the target.
- Treating an alarm as proof of a threat without following the secondary-check procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.